

WEB TECHNOLOGY

Academic Learning Module

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01 HTML Tutorial

Not Available

KEY TECHNICAL CONCEPTS

HTML (HyperText Markup Language) is the core language used to structure and display content on the web, defining elements like text, images, links, and page layout so browsers can render websites properly.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

- [Attributes Reference](#)
 - [Global Attributes Reference](#)
 - [Event Attributes Reference](#)
 - [DOM Reference](#)
 - [DOM Audio/Videos Reference](#)
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02 CSS Tutorial

Not Available

KEY TECHNICAL CONCEPTS

CSS stands for Cascading Style Sheets. It is a stylesheet language used to style and enhance website presentation. CSS is one of the three main components of a webpage, along with HTML and JavaScript.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

03 JavaScript Tutorial

Not Available

KEY TECHNICAL CONCEPTS

JavaScript is a programming language used to create dynamic content for websites. It is a lightweight, cross-platform, and single-threaded programming language. It's an interpreted language that executes code line by line, providing more flexibility.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

04 HTTP - Hypertext Transfer Protocol

Not Available

KEY TECHNICAL CONCEPTS

HTTP stands for Hypertext Transfer Protocol, and it's the system that allows communication between web browsers (like Google Chrome or Firefox) and web servers.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

05 Domain Name System (DNS)

Not Available

KEY TECHNICAL CONCEPTS

The Domain Name System (DNS) is a hierarchical and distributed naming system that translates human-readable domain names into IP addresses, enabling users to access websites easily.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

06 REST API Introduction

Not Available

KEY TECHNICAL CONCEPTS

A REST API (Representational State Transfer API) enables communication between client and server over HTTP. It exchanges data typically in JSON format using standard web protocols.

CODE IMPLEMENTATION

```
// Import the Express module const express = require('express'); const app = express(); const port = 3000; app.use(express.json()); // Define a route for GET requests app.get('/users', (req, res) => { res.json({ message: 'Returning list of users' }); }); // Define a route for POST requests app.post('/users', (req, res) => { const newUser = req.body; res.json({ message: 'User created', user: newUser }); }); // Define a route for PUT requests app.put('/users/:id', (req, res) => { const userId = req.params.id; const updatedUser = req.body; res.json({ message: `User with ID ${userId} updated`, updatedUser }); }); // Define a route for DELETE requests app.delete('/users/:id', (req, res) => { const userId = req.params.id; res.json({ message: `User with ID ${u
```

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

07 Not Available

Not Available

KEY TECHNICAL CONCEPTS

Not Available

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

08 TCP/IP Model

Not Available

KEY TECHNICAL CONCEPTS

The TCP/IP model is a layered networking framework that explains how data is communicated between devices over a network using standardized protocols to ensure reliable and efficient transmission.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Limited Multimedia Support:Original design focused on data, not optimized for real-time audio/video (needs extra protocols).

REFERENCES / RELATED LINKS

Not Available

09 Responsive Web Design

Not Available

KEY TECHNICAL CONCEPTS

Responsive Web Design ensures websites automatically adapt to different screen sizes, providing a smooth and user-friendly experience across all devices.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Google AI Studio (Voe2): Best Practices, Use Cases, Features5min read

REFERENCES / RELATED LINKS

Not Available

10 JSON Tutorial

Not Available

KEY TECHNICAL CONCEPTS

JSON (JavaScript Object Notation) is a widely-used, lightweight data format for representing structured data.

CODE IMPLEMENTATION

```
#include <iostream> #include <string> #include <nlohmann/json.hpp> using namespace std; using json = nlohmann::json; int main() { string json_s = R("{\"name\": \"Mohit\", \"age\": 30}"); json obj = json::parse(json_s); string name = obj["name"]; int age = obj["age"]; cout << "Name: " << name << ", Age: " << age << endl; return 0; }
```

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

11 Ajax Introduction

Not Available

KEY TECHNICAL CONCEPTS

Ajax (Asynchronous JavaScript and XML) is a powerful technique used in web development to enhance user experience by allowing pages to be updated asynchronously, without the need to refresh the entire page. This results in faster load times and more dynamic content, improving the overall functionality and responsiveness of web applications.

CODE IMPLEMENTATION

```
req.open('POST', 'file-url', true); req.setRequestHeader('Content-Type',  
'application/x-www-form-urlencoded'); req.send('name=value');
```

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available

12 WebSocket and Its Difference from HTTP

Not Available

KEY TECHNICAL CONCEPTS

HTTP and WebSocket are both ways for computers to talk to each other, but they work in different ways. HTTP is used for simple requests, like when you load a webpage, where the computer sends a request and the server replies, then the connection is closed. WebSocket keeps the connection open, allowing for real-time, two-way communication, making it great for things like live chats or online games where constant updates are needed.

CODE IMPLEMENTATION

Not Available

COMPLEXITY ANALYSIS

Not Available

REFERENCES / RELATED LINKS

Not Available